20 $f(x) = \dfrac{1-x}{1+x}$

(a) Show that $ff(x) = x$

Let $u = \dfrac{1-x}{1+x}$

$ff(x) = \dfrac{1-u}{1+u}$

1 mark
Compose f with itself.

Substitute u; multiply top and bottom by $(1+x)$.

1 mark
Clear both small denominators.

$\dfrac{1+x-(1-x)}{1+x+(1-x)} = \dfrac{2x}{2} = x$

1 mark
Simplify to x with full working.

Valid for $x \neq -1$.

(3)

(b) Hence, write down $f^{-1}(x)$

$f^{-1}(x) = \dfrac{1-x}{1+x}$

1 mark
Same function reverses f.

Since $f(f(x)) = x$, f is its own inverse.

(1)

(Total for Question 20 is 4 marks)